

Lab 12

```
#include <ESP8266WiFi.h>
#include <ESP8266HTTPClient.h>

#define LED D1

const char* ssid = "wifi name";
const char* password = "password";

String readAPIKey = "UCQLP46QUXWV9099";
String channelID = "3324687";

WiFiClient client;
HTTPClient http;

void setup() {
    Serial.begin(115200);
    pinMode(LED, OUTPUT);

    WiFi.begin(ssid, password);

    Serial.print("Connecting to WiFi");
    while (WiFi.status() != WL_CONNECTED) {
        delay(1000);
        Serial.print(".");
    }

    Serial.println("\nConnected!");
}

void loop() {
    if (WiFi.status() == WL_CONNECTED) {

        String url = "http://api.thingspeak.com/channels/" + channelID +
            "/fields/1/last.txt?api_key=" + readAPIKey;

        http.begin(client, url);
        int httpCode = http.GET();

        if (httpCode > 0) {
            String payload = http.getString();

            Serial.print("RAW: ");
            Serial.println(payload);

            payload.trim();

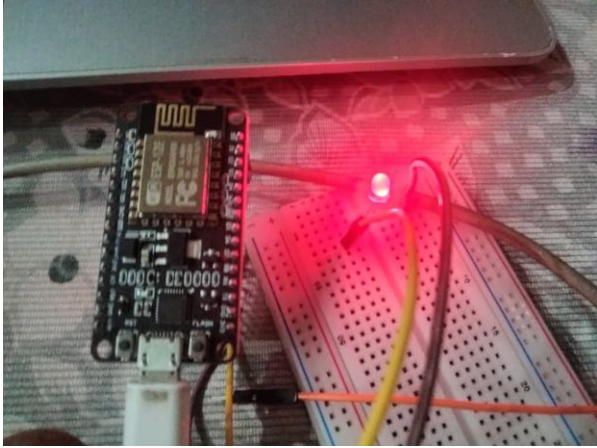
            if (payload == "1") {
                digitalWrite(LED, HIGH);
                Serial.println("LED ON");
            } else if (payload == "0") {
                digitalWrite(LED, LOW);
                Serial.println("LED OFF");
            } else {
                Serial.println("Invalid data");
            }
        }
    }
    else {
        Serial.print("HTTP Error: ");
        Serial.println(httpCode);
    }

    http.end();
}
```

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    }  
    delay(5000);  
}
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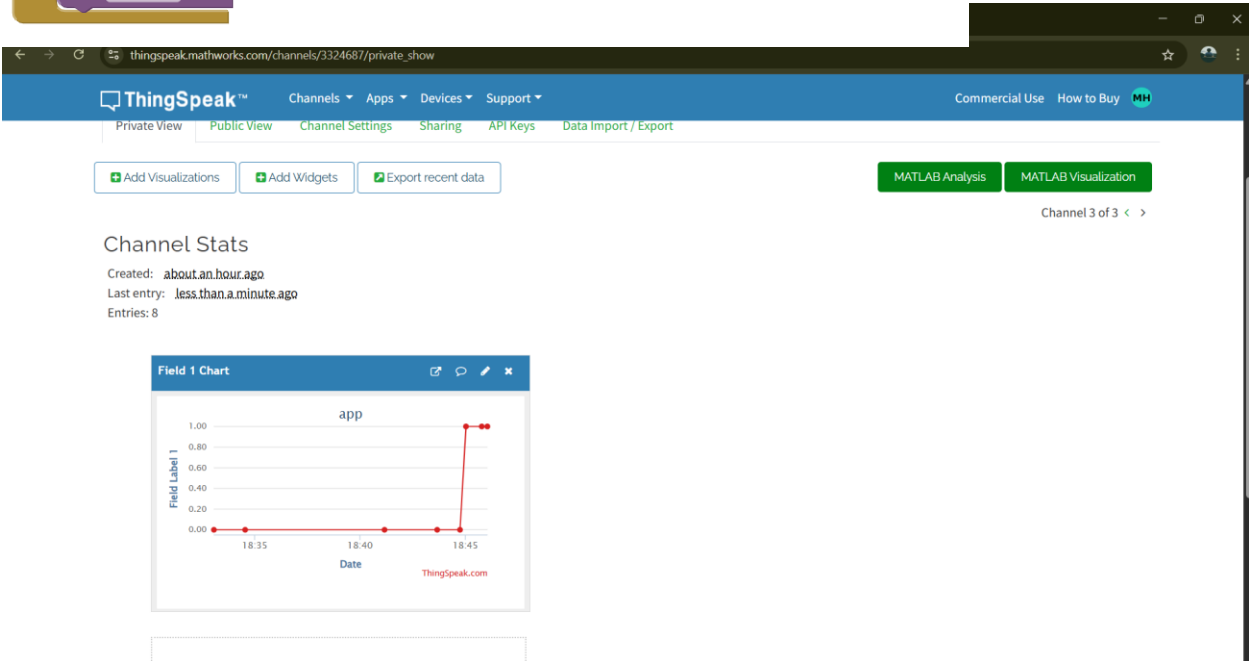
```
File Edit Search Tools Help
NodeMCU 1.0 (ESP-12)
watchdog_wdt

12 WiFiClient client;
13 WiFiClient http;
14
15 void setup() {
16   Serial.begin(115200);
17   pinMode(LED, OUTPUT);
18
19   WiFi.begin(ssid, password);
20
21   Serial.print("Connecting to WiFi");
22   while (WiFi.status() != WL_CONNECTED) {
23     delay(1000);
24     Serial.print(".");
25   }
26
27   Output - Serial Monitor X
28   Message (Enter to send message to NodeMCu 1.0 (ESP-12) Module on COM4)
29   Name List 115200 baud
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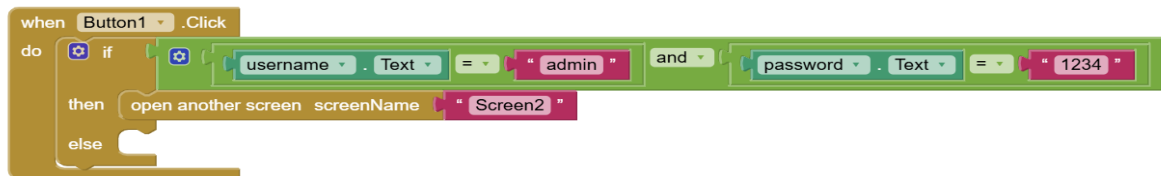
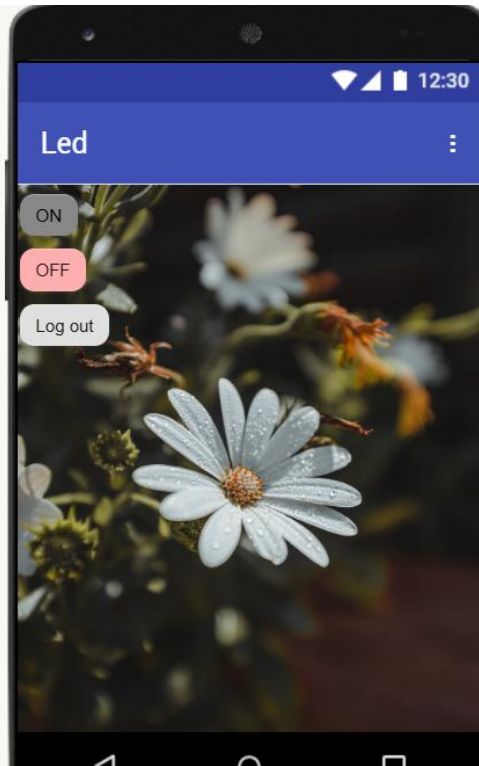
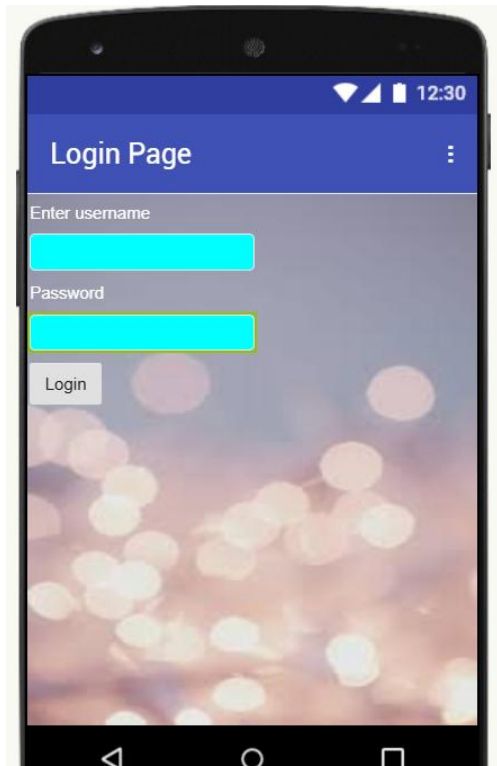


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when Button1 .Click
do
  set Web1 .Url to "sk.com/update?api_key=DL31YRMA25Y05F57&field1=0"
  call Web1 .Get
```

```
when Button2.Click
do
  set Web1.Url to "https://api.thingspeak.com/update?api_key=DL31YR..."
  call Web1.Get
```



Post lab



Screen 2

